

SUPERSTORE SALES ANALYSIS

SQL-Based Business Insights & Sales Performance



Presented by

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Tools: MySQL | Excel | Canva

Total Sales, Profit, Orders, Customers

SQL Query

```
SELECT
    ROUND(SUM(sales), 2) AS Total_sales,
    ROUND(SUM(profit), 2) AS Total_Profit,
    COUNT(DISTINCT (order_id)) AS Total_Orders,
    COUNT(DISTINCT (customer_id)) AS Total_Customers
FROM
    sales;
```

Result

Total_sales	Total_Profit	Total_Orders	Total_Customers
2297200.86	286397.79	5009	793

Sales and Profit changes month-by-month

SQL Query

```
SELECT
    YEAR(Order_Date) AS year,
    MONTH(order_date) AS month,
    ROUND(SUM(sales), 2) AS Sales,
    ROUND(SUM(profit), 2) AS Profit
FROM
    sales
GROUP BY YEAR(Order_Date) , MONTH(order_date)
ORDER BY year , month;
```

Result

year	month	Sales	Profit
2014	1	14236.90	2450.18
2014	2	4519.89	862.30
2014	3	55691.01	498.72
2014	4	28295.35	3488.86
2014	5	24929.28	2913.03
2014	6	30880.82	4675.78
2014	7	33946.39	-841.46
2014	8	27909.47	5318.11
2014	9	85491.65	8628.86
2014	10	31417.10	3437.86
2014	11	78628.72	9292.15
2014	12	68300.92	8819.67
2015	1	18174.08	-3280.97

Top 10 Products by sales

SQL Query

```
SELECT
    Product_name, ROUND(SUM(sales), 2) AS Total_sales
FROM
    sales
GROUP BY product_name
ORDER BY SUM(sales) DESC
LIMIT 10;
```

Result

Product_name	Total_profit
Canon imageCLASS 2200 Advanced Copier	25199.94
Fellowes PB500 Electric Punch Plastic Comb Bind...	7753.06
Hewlett Packard LaserJet 3310 Copier	6983.89
Canon PC1060 Personal Laser Copier	4570.94
HP Designjet T520 Inkjet Large Format Printer -...	4094.98
Ativa V4110MDD Micro-Cut Shredder	3772.95
3D Systems Cube Printer- 2nd Generation- Mag...	3717.97
Plantronics Savi W720 Multi-Device Wireless He...	3696.28
Ibico EPK-21 Electric Binding System	3345.29
Zebra ZM400 Thermal Label Printer	3343.53

Products with high sales but negative Profit

SQL Query

```
SELECT
    product_name,
    ROUND(SUM(sales), 2) AS Total_sales,
    ROUND(SUM(profit), 2) AS Total_profit
FROM
    sales
GROUP BY product_name
HAVING SUM(profit) < 0
ORDER BY Total_sales DESC
limit 10;
```

Result

product_name	Total_sales	Total_profit
Cisco TelePresence System EX90 Videoconferen...	22638.48	-1811.08
GBC DocuBind P400 Electric Binding System	17965.07	-1878.17
High Speed Automatic Electric Letter Opener	17030.31	-262.00
Lexmark MX611dhe Monochrome Laser Printer	16829.90	-4589.97
Martin Yale Chadless Opener Electric Letter Ope...	16656.20	-1299.19
Bretford Rectangular Conference Table Tops	12995.29	-327.25
Cubify CubeX 3D Printer Double Head Print	11099.96	-8879.97
Chromcraft Bull-Nose Wood Oval Conference T...	9917.64	-2876.11
Bush Advantage Collection Racetrack Conferen...	9544.73	-1934.40
Tennsco Single-Tier Lockers	9158.30	-825.76

Category and Sub-Category with most sales and profit

SQL Query

```
SELECT
    category,
    sub_category,
    ROUND(SUM(sales), 2) AS Total_sales,
    ROUND(SUM(profit), 2) AS Total_profit
FROM
    sales
GROUP BY category , sub_category
ORDER BY total_sales DESC , total_profit DESC;
```

Result

category	sub_category	Total_sales	Total_profit
Technology	Phones	330007.05	44516.25
Furniture	Chairs	328449.10	26590.15
Office Supplies	Storage	223843.61	21279.05
Furniture	Tables	206965.53	-17725.59
Office Supplies	Binders	203412.73	30221.64
Technology	Machines	189238.63	3384.73
Technology	Accessories	167380.32	41936.78
Technology	Copiers	149528.03	55617.90
Furniture	Bookcases	114880.00	-3472.56
Office Supplies	Appliances	107532.16	18138.07
Furniture	Furnishings	91705.16	13059.25
Office Supplies	Paper	78479.21	34053.34
Office Supplies	Supplies	46673.54	-1188.99
Office Supplies	Art	27118.79	6527.96
Office Supplies	Envelopes	16476.40	6964.10
Office Supplies	Labels	12486.31	5546.18
Office Supplies	Fasteners	3024.28	949.53

Customer Segment with highest Sales and Profit

SQL Query

```
SELECT
    segment,
    ROUND(SUM(sales), 2) AS Total_sales,
    ROUND(SUM(profit), 2) AS Total_profit
FROM
    sales
GROUP BY segment;
```

Result

segment	Total_sales	Total_profit
Consumer	1161401.35	134119.33
Corporate	706146.37	91979.45
Home Office	429653.15	60299.01

Regions and State with highest and lowest Profit

```
SELECT
    region, ROUND(SUM(profit), 2) AS total_profit
FROM
    sales
GROUP BY region
ORDER BY total_profit DESC;
```

SQL Query

```
SELECT
    state, ROUND(SUM(profit), 2) AS total_profit
FROM
    sales
GROUP BY state
ORDER BY total_profit DESC
LIMIT 10;
```

region	total_profit
West	108418.79
East	91522.84
South	46749.71
Central	39706.45

Result

state	total_profit
California	76381.60
New York	74038.64
Washington	33402.70
Michigan	24463.15
Virginia	18598.00
Indiana	18382.97
Georgia	16250.08
Kentucky	11199.70
Minnesota	10823.22
Delaware	9977.37

Discount affecting Profit

SQL Query

```
SELECT
    Discount,
    ROUND(SUM(sales), 2) AS Total_sales,
    ROUND(SUM(profit), 2) AS total_profit,
    ROUND(AVG(profit), 2) AS avg_profit
FROM
    sales
GROUP BY Discount
ORDER BY Discount;
```

Result

Discount	Total_sales	total_profit	avg_profit
0.00	1087908.47	320987.88	66.90
0.10	54369.35	9029.21	96.06
0.15	27558.52	1418.98	27.29
0.20	764594.37	90338.16	24.70
0.30	103226.66	-10369.34	-45.68
0.32	14493.46	-2391.16	-88.56
0.40	116417.78	-23057.08	-111.93
0.45	5484.97	-2493.12	-226.65
0.50	58918.54	-20506.51	-310.70
0.60	6644.70	-5944.64	-43.08
0.70	40620.28	-40075.46	-95.87
0.80	16963.76	-30539.13	-101.80

Ship mode with highest Profit

SQL Query

```
SELECT
    ship_mode, ROUND(SUM(sales), 2) AS Total_sales
FROM
    sales
GROUP BY ship_mode
ORDER BY SUM(sales) DESC;
```

Result

ship_mode	Total_sales
Standard Class	1358215.74
Second Class	459193.57
First Class	351428.42
Same Day	128363.13

Ranking Products within each Category

SQL Query

```
SELECT *
from(
SELECT
    category,
    product_name,
    round(sum(profit),2)as total_profit,
    rank() over(
        partition by category order by sum(profit) desc
    ) as rnk
    from sales
    group by category, product_name
) as RankedProducts
where rnk=1 ;
```

Result

	category	product_name	total_profit	rnk
►	Furniture	Hon Deluxe Fabric Upholstered Stacking Chairs-...	1707.87	1
	Office Supplies	Fellowes PB500 Electric Punch Plastic Comb Bind...	7753.06	1
	Technology	Canon imageCLASS 2200 Advanced Copier	25199.94	1

Key Findings...

- The Superstore generated **2.30 crore** in total sales and **28.63 lakhs** in profit from **5,009** orders placed by **793** customers.
- Sales and profit fluctuated across different months, indicating seasonal variations in business performance.
- A few products contributed the highest share of total sales, while some products generated high sales but negative profit, reducing overall profitability.
- Product categories and sub-categories showed different levels of sales and profit, with some categories performing much better than others.
- Customer segments contributed differently to sales and profit, showing that some customer groups were more valuable to the business.
- Higher discounts generally resulted in lower profit margins, affecting overall business profitability.

Business Recommendations

- Review products that generate high sales but negative profit and revise their pricing or discount strategy.
- Focus marketing efforts on the top-selling and high-profit products to increase revenue.
- Avoid offering excessive discounts, especially on products with already low profit margins.
- Analyze low-performing regions to identify opportunities for improving sales and profitability.
- Monitor monthly sales trends regularly to improve inventory planning and business decisions.